OSHA 29 CFR 1926 · SUBPART L · §1926.451 & §1926.452 — STEEL SCAFFOLDING ONLY

STEEL SCAFFOLD STANDARDS & MEASUREMENTS

Complete reference for **Steel Couplers, Steel Pipes/Tubes, Wire Rope, Platform, Guardrail, Access & Power Line Clearances**
Source: OSHA_29_CFR_1926.pdf (pp. 62–76) & CFR-2020-title29-vol8-part1926-subpartL.pdf · Aluminum, wood, and FRP sections removed

1 — SCAFFOLD CAPACITY REQUIREMENTS

§1926.451(a)

SCAFFOLD / COMPONENT TYPE	LOAD FACTOR REQUIRED	CONDITION / NOTES	REF.
General scaffold & all components	`≥ 4× maximum intended load`	Must support own weight PLUS 4× intended load without failure	§451(a)(1)
Roof/floor direct connections & counterweights (adjustable suspension)	`≥ 4× tipping moment at rated hoist load` OR `≥ 1.5× tipping moment at stall load` — whichever is greater	Take the larger value of both conditions	§451(a)(2)
Suspension rope + hardware — **non-adjustable** suspension scaffold	`≥ 6× maximum intended load`	Each rope individually must meet this factor	§451(a)(3)
Suspension rope + hardware — **adjustable** suspension scaffold	`≥ 6× maximum intended load` at rated hoist load OR `≥ 2× stall load` — whichever is greater	Both conditions must be checked; use the larger value	§451(a)(4)
Scaffold hoist stall load	`≤ 3× rated load`	Stall load must NOT exceed 3× the hoist's rated load	§451(a)(5)

i Scaffolds **shall be designed by a qualified person** and constructed/loaded in accordance with that design. Non-mandatory Appendix A provides example criteria.

2 — PLATFORM CONSTRUCTION MEASUREMENTS

§1926.451(b)

REQUIREMENT	MEASUREMENT / TOLERANCE	EXCEPTION / NOTES	REF.
Gap between adjacent platform units	≤ 1 inch (2.5 cm)	Larger gap allowed only if employer demonstrates necessity (e.g., around uprights with side brackets)	§451(b)(1)(i)
Gap between platform and uprights (when wider gap demonstrated necessary)	≤ 9½ inches (24.1 cm)	Platform must be planked as fully as possible before this exception applies	§451(b)(1)(ii)
Minimum platform / walkway width — general	≥ 18 inches (46 cm)	Applies to all scaffold platforms and walkways unless exceptions below	§451(b)(2)
Minimum width — ladder jack, top plate bracket, roof bracket, pump jack scaffolds	≥ 12 inches (30 cm)	No minimum width for boatswains' chairs	§451(b)(2)(i)
Front edge of platform from face of work — general	≤ 14 inches (36 cm)	If exceeded, guardrail and/or PFAS required along front edge	§451(b)(3)
Front edge from face — outrigger scaffolds	≤ 3 inches (8 cm)	Stricter limit for outrigger scaffolds only	§451(b)(3)(i)
Front edge from face — plastering & lathing operations	≤ 18 inches (46 cm)	More lenient limit for these specific operations only	§451(b)(3)(ii)
Platform end minimum overhang over support centerline	≥ 6 inches (15 cm)	Unless cleated, hooked, or restrained by equivalent means	§451(b)(4)
Maximum platform overhang — platforms ≤ 10 ft long	≤ 12 inches (30 cm)	Unless cantilevered portion designed to support load without tipping, or guardrails block access to cantilevered end	§451(b)(5)(i)
Maximum platform overhang — platforms > 10 ft long	≤ 18 inches (46 cm)	Same tipping / guardrail exceptions apply	§451(b)(5)(ii)
Platform overlap (to create long platform)	≥ 12 inches (30 cm)	Overlap must occur only over supports; less than 12 in. only if platforms restrained to prevent movement	§451(b)(7)
Maximum platform deflection under load	≤ 1/60 of span	Applies to all platforms when loaded	§451(f)(16)

! **Mixed manufacturers:** Steel scaffold components from different manufacturers shall NOT be intermixed unless they fit together without force and structural integrity is maintained. If modified to intermix, a competent person must verify structural soundness. **Dissimilar metals** must not be used together unless a competent person confirms galvanic action will not reduce strength below the §451(a)(1) requirement.

3 — SUPPORTED SCAFFOLD CRITERIA

§1926.451(c)

REQUIREMENT	MEASUREMENT / RULE	NOTES	REF.
Height-to-base-width ratio requiring tie/brace	`> 4:1`	Scaffolds exceeding this ratio MUST be restrained from tipping by guying, tying, bracing, or equivalent means	§451(c)(1)
Vertical spacing of ties/braces — scaffolds **≤ 3 ft wide**	`Every ≤ 20 ft (6.1 m)`	First tie at closest horizontal member to 4:1 height; repeated every 20 ft thereafter	§451(c)(1)(ii)
Vertical spacing of ties/braces — scaffolds **> 3 ft wide**	`Every ≤ 26 ft (7.9 m)`	First tie at closest horizontal member to 4:1 height; repeated every 26 ft thereafter	§451(c)(1)(ii)
Top tie/brace placement on completed scaffold	`≤ 4:1 height from top`	Topmost tie must be no further than the 4:1 height down from the top	§451(c)(1)(ii)
Horizontal intervals for ties/braces	`≤ 30 ft (9.1 m)`	Measured from one end toward the other (not from both ends simultaneously)	§451(c)(1)(ii)
Base plates and mud sills	Required under poles, legs, posts, frames, uprights	Footings shall be level, sound, rigid, capable of supporting loaded scaffold without settling. Unstable objects (barrels, boxes, loose brick) prohibited as supports.	§451(c)(2)

4 — SUSPENSION SCAFFOLD CRITERIA

§1926.451(d)

COMPONENT / REQUIREMENT	MEASUREMENT / STANDARD	NOTES	REF.
Support surface capacity (outrigger beams, cornice hooks, parapet clamps)	`≥ 4× load at rated hoist` OR `≥ 1.5× load at stall capacity` — whichever greater	Evaluated by a competent person before scaffold is used	§451(d)(1)
Outrigger beam — material	**Structural metal** or equivalent strength	Must be restrained to prevent movement	§451(d)(2)
Outrigger beam tiebacks — strength	Equivalent in strength to suspension ropes	Required where beams not stabilized by bolts/direct connections to floor/roof deck. **Single tiebacks installed at an angle — PROHIBITED**	§451(d)(3)(vi–x)
Tieback installation angle	Perpendicular to face of building OR opposing-angle tiebacks	Tiebacks anchored to structural members only — NOT standpipes, vents, piping, or electrical conduit	§451(d)(3)(ix–x)
Counterweight material	Non-flowable; specifically designed counterweights only	**Sand, gravel, masonry units, roofing felt — PROHIBITED.** Secured by mechanical means. Must NOT be removed until scaffold disassembled.	§451(d)(3)(ii–v)
Outrigger beam — stop bolts/shackles	Required at **BOTH** ends	Channel iron beams: flanges turned out when substituting for I-beams	§451(d)(4)(i–ii)
Outrigger beam — web orientation	Web in **vertical position**	All bearing supports perpendicular to beam center line	§451(d)(4)(iii–iv)
Shackle/clevis positioning on outrigger	Directly over **centerline of stirrup**	—	§451(d)(4)(v)
Cornice hooks / roof hooks / parapet clamps — material	**Steel** or wrought iron	Must be supported by bearing blocks; tiebacks equivalent in strength to hoisting rope	§451(d)(5)(i–iv)
Winding drum hoists — minimum rope wraps at lowest travel point	`≥ 4 wraps of suspension rope`	Other hoist types: rope long enough to lower scaffold without rope end passing through hoist	§451(d)(6)
Repaired wire rope as suspension rope	**PROHIBITED**	Wire ropes may not be joined except via eye-splice thimbles with shackles or coverplates + bolts	§451(d)(7–8)
Welding — insulation coverage above hoist	`≥ 4 ft (1.2 m)` of insulating material above hoist	Required when welding from suspended scaffolds to prevent current arcing through wire rope. Hoists must be covered with insulated protective covers.	§451(f)(17)(ii–iii)
Gasoline-powered hoists on suspension scaffolds	**PROHIBITED**	Power-operated hoist gears and brakes must be enclosed	§451(d)(14–15)

5 — WIRE ROPE & CLIP REQUIREMENTS

§1926.451(d)(10–12)

WIRE ROPE REPLACEMENT CRITERIA — REPLACE IF ANY EXIST

Broken wires — random distribution	≥ 6 wires in one rope lay
Broken wires — one strand, one lay	≥ 3 wires
Diameter loss (abrasion/ corrosion/ peening)	> 1/3 of original outside wire diameter
Kinks impairing tracking/wrapping	REPLACE
Any physical damage impairing strength	REPLACE
Heat damage (torch or electrical wires)	REPLACE
Evidence secondary brake activated (overspeed)	REPLACE

WIRE ROPE CLIP REQUIREMENTS — §451(D)(12)

Minimum number of clips	≥ 3 clips
Minimum spacing between clips	≥ 6 rope diameters apart
Installation standard	Per manufacturer's recommendations
Retightening — after initial loading	**Required** per manufacturer
Retightening — each workshift start	**Inspect & retighten**
U-bolt clips at suspension hoist point	**PROHIBITED**
U-bolt placement when used elsewhere	U-bolt over *dead end*; saddle over *live end*

! Ropes shall be **inspected by a competent person before each workshift** and after every occurrence that could affect rope integrity. Swaged attachments or spliced eyes shall not be used unless made by the wire rope manufacturer or a qualified person.

6 — TUBE & COUPLER SCAFFOLD — SPECIFIC REQUIREMENTS

§1926.452(b) + Appendix A(b)

ℹ These requirements apply *in addition to* all general requirements in §1926.451. Tube and coupler scaffolds over **125 feet** in height must be designed by a **registered professional engineer**.

COMPONENT / REQUIREMENT	MEASUREMENT / STANDARD	NOTES	REF.
Coupler material — **PERMITTED**	Drop-forged **steel**, malleable iron, or structural-grade steel	**Gray cast iron — PROHIBITED**	§452(b)(9)
Height limit without PE design	`≤ 125 ft (38.1 m)`	Above 125 ft requires design by registered professional engineer	§452(b)(10)
Transverse bracing ("X" pattern)	At both scaffold ends + at least every **3rd set of posts** horizontally; every **4th runner** vertically	Building ties installed at bearer levels between transverse bracing per §451(c)(1)	§452(b)(2)
Longitudinal bracing angle	`45° (± 5°)`	From base of end posts to top. Repeated at ≤ every 5th post if scaffold length > height; alternating directions if length < height	§452(b)(3)
Bearer-to-runner coupler position	Inboard coupler must bear **directly on the runner coupler**	When bearers are coupled to runners, couplers shall be as close to posts as possible	§452(b)(5)
Bearer extension beyond post/runner	Must extend beyond posts and runners with **full contact with coupler**	—	§452(b)(6)
Runner placement	On **both inside and outside posts** at level heights	Steel tube and coupler guardrails/midrails on outside posts may substitute for outside runners	§452(b)(7)
Runners on straight runs	Interlocked to form continuous lengths; coupled to **each post**	Bottom runners and bearers located as close to base as possible	§452(b)(8)
Bracing where post attachment not possible	Attach to runners **as close to post as possible**	—	§452(b)(4)

Appendix A — Steel Tube & Coupler Minimum Member Sizes

Non-mandatory Appendix A(b) · All members are steel tube/pipe

COMPONENT	LIGHT DUTY (25 LB/FT²)	MEDIUM DUTY (50 LB/FT²)	HEAVY DUTY (75 LB/FT²)
Posts, Runners & Braces	`Nominal 2 in. OD steel tube/pipe (1.90 in. actual OD)`	`Nominal 2 in. OD steel tube/pipe (1.90 in. actual OD)`	`Nominal 2 in. OD steel tube/pipe (1.90 in. actual OD)`
Bearers — post spacing 4 ft × 10 ft (light) / 4 ft × 7 ft (medium) / 6 ft × 6 ft (heavy)	`Nominal 2 in. OD (1.90 in.)`	`Nominal 2 in. OD (1.90 in.)`	`Nominal 2½ in. OD (2.375 in.)`
Bearers — post spacing 6 ft × 8 ft (light) / 4 ft × 7 ft alt. (medium)	`Nominal 2 in. OD (1.90 in.)`	`Nominal 2½ in. OD (2.375 in.)`	—
Maximum Runner Spacing — Vertical	`6 ft 6 in.`	`6 ft 6 in.`	`6 ft 6 in.`
Longitudinal diagonal bracing angle	`45° (± 5°)`		

** Bearers shall be installed in the direction of the shorter post-spacing dimension.*

Appendix A — Maximum Number of Planked Levels (Steel Tube & Coupler)

WORKING LEVELS	LIGHT DUTY — ADDITIONAL PLANKED LEVELS	MEDIUM DUTY — ADDITIONAL PLANKED LEVELS	HEAVY DUTY — ADDITIONAL PLANKED LEVELS	MAX. SCAFFOLD HEIGHT
1	16	11	6	125 ft
2	11	1	0	125 ft
3	6	0	0	125 ft
4	1	0	0	125 ft

7 — ACCESS REQUIREMENTS (LADDERS, STAIRS, RAMPS)

§1926.451(e)

ACCESS TYPE / REQUIREMENT	MEASUREMENT	NOTES	REF.
Trigger height requiring formal access provision	> 2 ft (0.6 m) above/below access point	**Crossbraces — PROHIBITED as means of access**	§451(e)(1)
Hook-on / attachable ladder — bottom rung above supporting level	≤ 24 in. (61 cm)	Must be specifically designed for the type of scaffold used	§451(e)(2)(ii)
Hook-on / attachable ladder — rest platforms (scaffold > 35 ft)	Every ≤ 35 ft (10.7 m) vertically	Required on supported scaffolds more than 35 ft high	§451(e)(2)(iii)
Hook-on / attachable ladder — minimum rung length	≥ 11½ in. (29 cm)	Uniformly spaced	§451(e)(2)(v)
Hook-on / attachable ladder — maximum rung spacing	≤ 16¾ in.	—	§451(e)(2)(vi)
Stairway-type ladder — bottom step above supporting level	≤ 24 in. (61 cm)	—	§451(e)(3)(i)
Stairway-type ladder — rest platforms	Every ≤ 12 ft (3.7 m) vertically	—	§451(e)(3)(ii)
Stairway-type ladder — minimum step width (standard)	≥ 16 in. (41 cm)	Mobile scaffold stairway-type: min. 11½ in. (30 cm)	§451(e)(3)(iii)
Stairtower — bottom step above supporting level	≤ 24 in. (61 cm)	—	§451(e)(4)
Stairtower stairrail height range	28 in. (71 cm) min – 37 in. (94 cm) max	Measured from upper surface of stairrail to tread surface, in line with riser face at tread's forward edge	§451(e)(4)(vii)
Stairtower — landing platform minimum size	≥ 18 in. × 18 in. (45.7 cm × 45.7 cm)	Required at each level	§451(e)(4)(viii)
Stairtower — stairway width between stairrails	≥ 18 in. (45.7 cm)	—	§451(e)(4)(ix)
Stairtower — stairway installation angle	40° – 60° from horizontal	—	§451(e)(4)(xi)
Stairtower — riser height variation tolerance per flight	≤ ¼ in. (0.6 cm)	Greater variation allowed only for top and bottom steps of entire system (not per flight)	§451(e)(4)(xiii)
Stairtower — tread depth variation tolerance per flight	≤ ¼ in.	—	§451(e)(4)(xiv)
Ramp/walkway — guardrail trigger height	≥ 6 ft (1.8 m) above lower levels	Guardrail system required at or above this height	§451(e)(5)(i)
Ramp/walkway — maximum slope	≤ 1:3 (20°)	—	§451(e)(5)(ii)
Ramp/walkway cleats — when slope steeper than 1:8	Cleats spaced ≤ 14 in. (35 cm) apart	Securely fastened to planks to provide footing	§451(e)(5)(iii)
Steel prefabricated access frame — minimum rung length	≥ 8 in. (20 cm)	If rungs < 11½ in.: fall protection / positioning device per §1926.502 required	§451(e)(6)(ii–iii)

ACCESS TYPE / REQUIREMENT	MEASUREMENT	NOTES	REF.
Steel prefabricated access frame — maximum rung spacing	`≤ 16¾ in. (43 cm)`	Non-uniform spacing from joining end frames allowed if resulting spacing ≤ 16¾ in.	§451(e)(6)(vi)
Steel prefabricated access frame — rest platforms (if > 35 ft)	`Every ≤ 35 ft (10.7 m)`	—	§451(e)(6)(v)
Direct access (no ladder) — max. horizontal distance to other surface	`≤ 14 in. (36 cm)`	Both horizontal AND vertical limits must be met simultaneously	§451(e)(8)
Direct access — max. vertical distance to other surface	`≤ 24 in. (61 cm)`	—	§451(e)(8)
Tubular welded frame — end frames as climbing devices	Horizontal members parallel, level, `≤ 22 in. apart vertically`	**Cross braces on tubular welded frame — PROHIBITED as access/egress**	§451(e)(9)(iii–iv)
Handrail clearance from other objects	`≥ 3 in. (7.6 cm)`	Applies to handrails and toprails used as handrails	§451(e)(4)(vi)

8 — GUARDRAIL SYSTEM REQUIREMENTS

§1926.451(g)(1) & (g)(4)

GUARDRAIL COMPONENT / REQUIREMENT	MEASUREMENT / STANDARD	NOTES	REF.
Fall protection trigger height	`> 10 ft (3.1 m)` above lower level	Each employee above this height must be protected from falling	§451(g)(1)
Toprail height — scaffolds placed in service **after Jan 1, 2000**	`38 in. – 45 in. (0.97 m – 1.2 m)`	May exceed 45 in. if conditions warrant and all other criteria are met	§451(g)(4)(ii)
Toprail height — scaffolds in service **before Jan 1, 2000** + all suspended scaffolds requiring both guardrail and PFAS	`36 in. – 45 in. (0.9 m – 1.2 m)`	—	§451(g)(4)(ii)
Midrail position	Approximately **midway** between toprail top edge and platform surface	Installed between toprail and platform	§451(g)(4)(iii–iv)
Intermediate members (balusters/additional rails) maximum spacing	`≤ 19 in. (48 cm) apart`	—	§451(g)(4)(vi)
Steel toprail load capacity — single-point & two-point adjustable suspension	`≥ 100 lb (445 N)` in any downward or horizontal direction	—	§451(g)(4)(vii)
Steel toprail load capacity — all other scaffolds	`≥ 200 lb (890 N)` in any downward or horizontal direction	—	§451(g)(4)(vii)
Steel midrail load capacity — 100 lb toprail systems	`≥ 75 lb (333 N)`	Force applied in any downward or horizontal direction at any point	§451(g)(4)(ix)
Steel midrail load capacity — 200 lb toprail systems	`≥ 150 lb (666 N)`	Force applied in any downward or horizontal direction at any point	§451(g)(4)(ix)
Steel crossbracing as midrail — crossing point height	`20 in. – 30 in. (0.5 m – 0.8 m)` above platform	Acceptable in place of midrail when crossing point is within this range	§451(g)(4)(xv)
Steel crossbracing as toprail — crossing point height	`38 in. – 48 in. (0.97 m – 1.3 m)` above platform	End points at each upright: `≤ 48 in. (1.3 m) apart`	§451(g)(4)(xv)
Guardrail on walkway within scaffold	Within `9½ in. (24.1 cm)` of and along at least one side	Minimum 200 lb toprail capacity required	§451(g)(1)(v)
Prohibited toprail/midrail materials	**Steel or plastic banding — PROHIBITED**	Despite being steel, banding does not meet structural requirements for toprail/midrail use	§451(g)(4)(xiii)

Fall Protection by Scaffold Type

SCAFFOLD TYPE	REQUIRED FALL PROTECTION	REF.
Boatswains' chair, catenary, float, needle beam, ladder jack	PFAS ONLY	§451(g)(1)(i)
Single-point or two-point adjustable suspension	PFAS + GUARDRAIL — BOTH required	§451(g)(1)(ii)
Self-contained adjustable — frame-supported	GUARDRAIL (200 LB MIN.)	§451(g)(1)(iv)
Self-contained adjustable — rope-supported	PFAS + GUARDRAIL (200 LB MIN.) — BOTH required	

SCAFFOLD TYPE	REQUIRED FALL PROTECTION	REF.
		§451(g)(1)(iv)
Overhand bricklaying from supported scaffold	PFAS OR GUARDRAIL (200 LB MIN.) — all open sides/ends except wall side	§451(g)(1)(vi)
All other scaffolds (general — steel frame, tube & coupler, system scaffold, etc.)	PFAS OR GUARDRAIL PER §451(G)(4)	§451(g)(1)(vii)

Appendix A — Steel Guardrail Member Minimum Strength Equivalents

MEMBER	STEEL ANGLE IRON	STEEL TUBE	POST SPACING (MAX.)
Toprails	1¼ in. × 1/8 in. structural angle	1 in. × 0.070 in. wall steel tube	≤ 8 ft
Midrails	1¼ × 1¼ × 1/8 in. structural angle	1 in. × 0.070 in. wall steel tube	
Toeboards	1¼ × 1¼ in. structural angle	1 in. × 0.070 in. wall steel tube	
Posts	1¼ × 1¼ × 1/8 in. structural angle	1 in. × 0.070 in. wall steel tube	
Overhead protection	Steel mesh screen: No. 18 gauge U.S. Standard wire, 1 in. mesh (between toeboard and midrail/toprail)		

9 — POWER LINE CLEARANCES

§1926.451(f)(6)

LINE TYPE	VOLTAGE	MINIMUM CLEARANCE	ALTERNATIVE CLEARANCE
Insulated Lines	Less than 300 V	3 ft (0.9 m)	—
	300 V to 50 kV	10 ft (3.1 m)	—
	More than 50 kV	10 ft (3.1 m) + 0.4 in. per kV over 50 kV	2× line insulator length, never less than 10 ft
Uninsulated Lines	Less than 50 kV	10 ft (3.1 m)	—
	More than 50 kV	10 ft (3.1 m) + 0.4 in. per kV over 50 kV	2× line insulator length, never less than 10 ft

✓ **Exception:** Scaffolds may be closer only when work requires it AND the utility company has been notified AND the utility has de-energized, relocated, or installed protective coverings on the lines.

10 — TOEBOARD REQUIREMENTS

§1926.451(h)(4)

TOEBOARD DIMENSIONS & TOLERANCES

Minimum toeboard height	≥ 3½ in. (9 cm)
Maximum clearance above walking surface	≤ ¼ in. (0.7 cm)
Maximum opening size (if not solid)	≤ 1 in. (2.5 cm) greatest dimension
Position on platform	Outermost edge; securely fastened
Required when platform above lower levels	> 10 ft (3.1 m)

TOEBOARD LOAD CAPACITY

Minimum force resistance	≥ 50 lb (222 N)
Direction of force	Any downward or horizontal direction
Application point	At any point along the toeboard

! **Absolute Prohibitions — Quick Reference (§1926.451)**

- Shore or lean-to scaffolds — use PROHIBITED (f)(2)
- Gray cast iron couplers — PROHIBITED (§452(b)(9))
- Steel or plastic banding as toprail or midrail — PROHIBITED (g)(4)(xiii)
- Crossbraces as means of access — PROHIBITED (e)(1)
- Cross braces on tubular welded frame scaffolds as access/egress — PROHIBITED (e)(9)(iv)
- Working on scaffolds during storms/high winds without competent person approval + PFAS or windscreens — PROHIBITED (f)(12)
- Makeshift devices (boxes, barrels) to increase working level height — PROHIBITED (f)(14)
- Ladders on scaffolds to increase working height (except large-area scaffolds meeting all criteria) — PROHIBITED (f)(15)
- Debris accumulation on platforms — PROHIBITED (f)(13)
- Gasoline-powered equipment/hoists on suspension scaffolds — PROHIBITED (d)(14)
- Repaired wire rope as suspension rope — PROHIBITED (d)(7)
- U-bolt clips at point of suspension for any scaffold hoist — PROHIBITED (d)(12)(v)
- Single tiebacks installed at an angle — PROHIBITED (d)(3)(x)

Source: OSHA 29 CFR 1926 Subpart L — §§1926.451, 1926.452, Appendix A & E · Steel scaffolding requirements only · Aluminum, wood, and FRP provisions have been removed · Always verify against current official CFR text at ecfr.gov